NETWORKING ASSIGNMENT

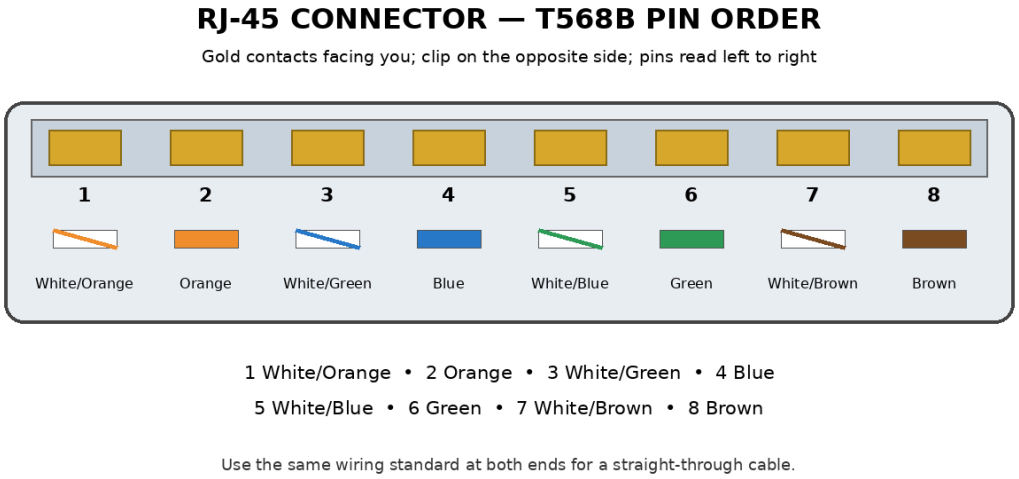
Ethernet Cables, T568B Wiring and RJ-45 Crimping

TASK 1: Cat6 Cable and T568B Pin Order

QUESTION

Draw a labeled Cat6 Ethernet cable end. Show all eight wire colours in the T568B order and label RJ-45 pins 1 through 8.

ANSWER



1 White/Orange • 2 Orange • 3 White/Green • 4 Blue
5 White/Blue • 6 Green • 7 White/Brown • 8 Brown

Use the same wiring standard at both ends for a straight-through cable.

Pin	Wire colour	Pin	Wire colour
1	White/Orange	5	White/Blue
2	Orange	6	Green
3	White/Green	7	White/Brown
4	Blue	8	Brown

The RJ-45 plug is commonly called an 8P8C connector. For a straight-through patch cable, use T568B at both ends.

TASK 2: Cable Types and Real-World Uses

QUESTION

List three network cable types. For each, name one real-world device or application and explain why that cable is suitable.

ANSWER

Cable type	Example use	Reason
Cat5e Ethernet	Home Wi-Fi router to desktop PC	Affordable and commonly supports Gigabit Ethernet over standard runs.
Cat6 Ethernet	Gaming console or POS terminal connected to a switch	Reliable wired connection with good performance and low latency.
Fiber optic	IPL streaming server or link between switches	High bandwidth over longer distances and immunity to electromagnetic interference.

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TASK 3: RJ-45 Crimping Procedure

QUESTION

Describe the process of crimping an RJ-45 plug onto an Ethernet cable in your own words. Include two mistakes that should be avoided.

ANSWER

1. Prepare the Cat5e/Cat6 cable, compatible RJ-45 plug, crimping tool, cable stripper/cutter, and cable tester.

2. Cut the cable end square, then carefully remove a short section of the outer jacket without cutting the inner conductors.

3. Untwist and straighten the eight wires only as much as necessary.

4. Arrange the wires in T568B order: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.

5. Hold the wires in order and trim the ends evenly.

6. Push all wires fully into the plug, making sure the cable jacket reaches the plug's strain-relief section.

7. Insert the plug into the crimping tool and squeeze firmly to secure the contacts.

8. Crimp the other end using the same standard for a straight-through cable. Test the cable with a LAN tester.

Two mistakes to avoid: (1) incorrect wire colour order, which can cause failed or unreliable connectivity; (2) wires not pushed fully into the plug or the outer jacket not held by the plug, causing a poor connection.

Video reference: Search YouTube for "How to Crimp RJ45 Connector".

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TASK 4: Cable Choices for an IPL Screening Gaming Zone

QUESTION

You are setting up a gaming zone for an IPL screening event with several consoles, screens, and a streaming system. Choose appropriate cable and connector types and explain your choices.

ANSWER

Use Cat6 Ethernet cable with RJ-45 connectors to connect consoles, the streaming computer, and network switches. Cat6 provides a fast, stable wired network and helps reduce latency during multiplayer gaming and streaming. Use a gigabit or faster Ethernet switch to connect devices and a router to provide internet access. If a network link must travel a long distance between rooms or buildings, use fiber-optic cable with compatible optical transceivers for the backbone.

For the actual video signal from each console or streaming computer to its screen, use HDMI cables. Ethernet connects devices to the network; it does not normally replace the direct HDMI display connection.

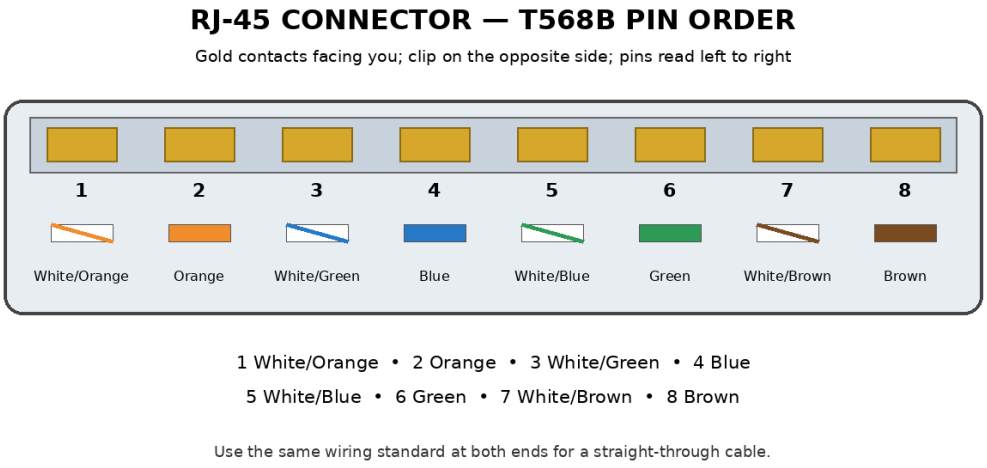
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TASK 5: Identify a Network Cable in Use

QUESTION

Provide a photo of a network cable or connector in use and describe the cable type and the devices it connects.

ANSWER



Reference description: This diagram shows an Ethernet cable's RJ-45 connector. In a typical home setup, an Ethernet patch cable connects a router's LAN port to a desktop computer, network switch, smart TV, or other network device.

For a photo of a real installation, take a picture of the cable plugged into your home or college router. Identify both ends of the cable and name the devices connected. If using an online photo, cite its source and make sure it clearly shows the cable or connector in use.

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